

GNN-Based BERT for Understanding Context from Music

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Abstract—Music is an inherently multi-layered acoustic signal whose semantic context spans melody, harmony, instrumentation, and relational chord progressions. We investigate a framework combining Graph Neural Networks (GNNs) on music structure graphs with Bidirectional Encoder Representations from Transformers (BERT) on natural-language descriptions. The FMA-medium audio pipeline successfully processes 24,980 of 25,000 tracks, excluding 20 decoding failures, under artist-disjoint partitions. GPU feature extraction and float16 memory-mapped storage enable efficient graph construction and training. The GNN study reports 43 experimental runs across five ablation axes: topology, node granularity, features, encoder, and depth. On the top-20 multi-label genre task, the selected GAT achieves 0.3544 Macro-F1 and 0.3673 AUC-PR, compared with 0.4061 and 0.4260 for a controlled CNN baseline. A separate BERT baseline predicts 50 MusicCaps aspect tags; these scores are not directly comparable with FMA genre scores. Fusion and contrastive dual-encoder results are retained as reported experiments, but their audio-text pairing and evaluation provenance remain undocumented in the supplied material. Consequently, they do not yet establish verified cross-modal alignment or retrieval performance.

Index Terms—graph neural networks, BERT, music information retrieval, multimodal fusion, contrastive learning, cross-attention

I. INTRODUCTION AND MOTIVATION

A single musical clip conveys multi-faceted attributes: genre, instrumentation, emotional valence, and harmonic progressions. Convolutional audio representations model local spectrotemporal patterns, whereas graph representations explicitly connect recurring musical events, temporal windows, and chords. Temporal and self-similarity edges offer a complementary representation of compositional relationships. Contextual language models such as BERT [3] provide linguistic representations of genre, style, and semantic description.

The CSE425 Supervised Neural Network project comprises audio preprocessing on FMA-medium [1], a multi-label BERT baseline, structural GNN encoders and ablations, GNN-BERT fusion, and a contrastive dual encoder for text-audio retrieval. This report distinguishes the documented standalone benchmarks from multimodal experiments whose pairing records are still required for validation.

II. DATASET AND AUDIO PIPELINE

A. Dataset Organization

FMA-medium contains 25,000 approximately 30-second tracks. The pipeline converts audio to 22,050 Hz mono and extracts 128-bin log-mel spectrograms and 12-bin chroma STFT features. Per-track zero-mean, unit-variance normalization preserves relative intra-clip dynamics (Fig. 1). The BERT pipeline uses MusicCaps [6], containing 5,521 music clips with expert captions and descriptive aspect tags. Its `aspect_list` field supplies a 50-label multi-hot vocabulary of frequent tags, including low quality, instrumental, emotional, noisy, passionate, and medium tempo. MusicCaps caption tagging and FMA genre prediction are separate tasks.

B. MusicCaps Data Partitioning

MusicCaps was divided into reproducible 70/15/15 train/validation/test partitions using a fixed random seed, yielding 3,864, 828, and 829 clips. The corresponding `train.jsonl`, `val.jsonl`, and `test.jsonl` files support consistent evaluation. Distinct YouTube video identifiers do not establish artist-disjointness; no MusicCaps artist-level overlap audit is supplied, so artist leakage cannot be ruled out.

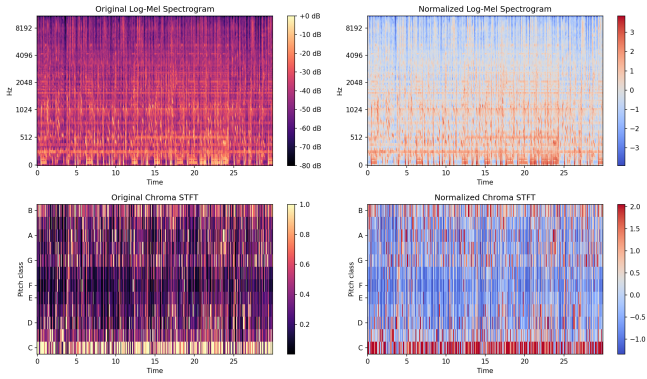


Fig. 1. Extracted and normalized 128-bin log-mel spectrogram and 12-bin chroma features.

C. Segmentation and Leakage Prevention

Each decoded track is initially partitioned into six approximately five-second segments, yielding mel tensors of shape $6 \times 128 \times 215$ and chroma tensors of shape $6 \times 12 \times 215$. The original FMA partitions contain 19,922 training, 2,505 validation, and 2,573 test tracks, totaling 25,000 *before filtering*. Twenty tracks failed decoding because of truncated MP3 bitstreams, including identifiers 099134, 108925, and 133297. Failures were catalogued in `results/extraction_failures.json`; `features_store.usable_ids()` excludes them downstream. Thus, 24,980 denotes successfully decoded tracks, not the sum of effective task-specific split sizes.

The primary GNN task uses the top-20 frequent genres from FMA’s `genres_all` metadata, multi-hot targets $\mathbf{y} \in \{0,1\}^{20}$, and BCEWithLogitsLoss. The contribution report gives 18,294 usable training tracks for this formulation. Effective validation and test counts after decoding and task filtering are not reported; the original 2,505 and 2,573 must not be substituted for them. Recovery requires filtered split manifests. A separate 16-class single-label task uses `genre_top` and cross-entropy loss.

The reported FMA artist audit found 4,463 training, 588 validation, and 593 test artists, with zero shared artist identifiers in every pair of partitions. Split metadata was augmented with `genres_all` and `artist_id` while preserving existing `genre` and `subset` fields. This audit concerns FMA only.

D. GPU Feature Extraction and Validation

The initial CPU implementation used `librosa` [7]. The contribution report records native failures in tuning estimation called by `chroma_stft`, causing multiprocessing workers to terminate, and a projected CPU runtime of approximately 35 minutes. Person 3 implemented `scripts/extract_features_gpu.py`: CPU `soundfile` decodes MP3s; PyTorch and `torchaudio` perform GPU resampling, STFT with FFT/window length 2,048 and hop 512, 128-bin mel filtering, and power-to-decibel conversion. A 12-bin chroma filterbank projects the power spectrum, followed by per-frame L_∞ normalization, bypassing

the failing tuning-estimation routine. Track normalization and segmentation follow extraction.

Comparison with non-crashing CPU outputs yielded Pearson correlations of 0.9950 for log-mel and 0.9999 for chroma. GPU extraction was reported at approximately 1.5 minutes for the corpus (approximately 272 tracks/s). The $23.3\times$ speedup is relative to the *projected* 35-minute CPU runtime, not a measured complete CPU-versus-GPU benchmark.

E. Memory-Mapped Feature Store

Person 3 replaced individual compressed archives with contiguous float16 NumPy memory-mapped arrays in `data/processed/store/`. The mel and chroma stores have allocated shapes (25000, 6, 128, 215) and (25000, 6, 12, 215), occupying approximately 8.2 GB and 0.8 GB. A JSON index maps track IDs to rows; allocated capacity does not imply that failed tracks are usable. Reported total storage fell from approximately 72 GB to 9 GB. Operating-system paging permits on-demand slices without loading entire arrays or opening thousands of archives.

III. BASELINE ARCHITECTURES AND TASK 1: BERT BASELINE

A. Random Baseline (B1)

A label-prior predictor uses training-label frequencies, achieving 0.0977 Macro-F1, 0.2599 Micro-F1, and 0.0821 AUC-PR on the FMA top-20 multi-label task.

B. CNN Baseline (B2)

Person 1’s four-block 2D CNN uses BatchNorm, MaxPool2d, and AdaptiveAvgPool2d on segmented mel spectrograms. Person 3 replicated it under the GNN’s multi-label partitions, validation-threshold procedure, and test metrics, obtaining 0.4061 Macro-F1, 0.5032 Micro-F1, and 0.4260 AUC-PR. The separate 16-class experiment is reported in Section V.

C. BERT Multi-Label Classifier (B3 / Task 1)

The `bert-base-uncased` encoder [3] is fine-tuned on MusicCaps captions tokenized to at most 128 tokens. The 768-dimensional [CLS] representation feeds a linear head producing 50 tag logits with independent sigmoid probabilities. All parameters are unfrozen and trained using binary cross-entropy and AdamW with learning rate 2×10^{-5} and batch size 16 for six epochs; checkpoint selection uses validation Macro-F1.

Reported *validation* Macro-F1 rises from 0.1084 to 0.6160 and Micro-F1 from 0.2906 to 0.7273. Figure 2 shows the training and validation loss curves.

The final BERT results (Macro-F1 0.6126, Micro-F1 0.7232, AUC-PR 0.7130) were computed by `evaluate.py` on the held-out test split (`test.jsonl`, 829 clips), using the epoch-6 checkpoint selected by highest validation Macro-F1. The test split was not used during training or validation, including checkpoint selection. These are *test-set scores*, distinct from the per-epoch validation Macro-F1 and Micro-F1 results.

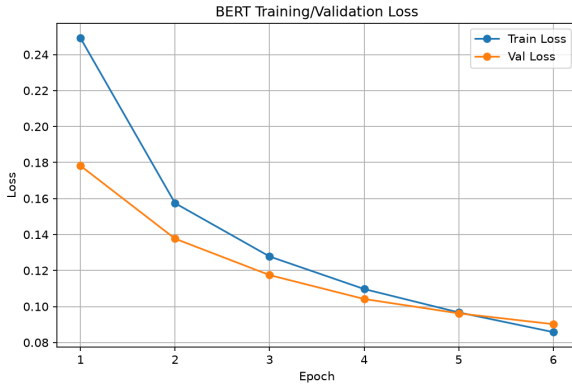


Fig. 2. Task 1 BERT multi-label classifier training and validation loss curves.

TABLE I
QUALITATIVE PREDICTIONS OF THE BERT CLASSIFIER ON MUSICCAPS.

Caption (excerpt)	True tags	Predicted tags
This music is instrumental. The tempo is slow with an electronic keyboard...	instrumental, medium tempo, amateur	instrumental, slow tempo
Low Quality recording featuring a person tuning...	low quality	low quality
Heavily processed girl band... pop or R&B style...	pop	none
This is a drumline stick trick recording...	none	none

Qualitative examples in Table I suggest stronger performance on explicit descriptors than on indirect stylistic cues.

IV. TASK 2: MUSIC STRUCTURE GRAPHS AND GNN ENCODERS

A. Graph Construction

1) *Segment graphs*: The stored features are stitched into 1,290 frames and re-sliced into $N \in \{6, 15, 30, 60\}$ windows without decoding audio again. Node descriptors concatenate mel/chroma means and standard deviations:

$$\mathbf{x}_i = [\boldsymbol{\mu}_{\text{mel}} \parallel \boldsymbol{\mu}_{\text{chroma}} \parallel \boldsymbol{\sigma}_{\text{mel}} \parallel \boldsymbol{\sigma}_{\text{chroma}}] \in \mathbb{R}^{280}. \quad (1)$$

The component dimensions are 128, 12, 128, and 12. Four segment-edge modes are evaluated:

- 1) **Temporal**: connect consecutive windows bidirectionally with unit weights.
- 2) **kNN**: preserve temporal edges and add each node's $k = 3$ most cosine-similar non-adjacent neighbors, with similarity weights.
- 3) **Threshold**: preserve temporal edges and add non-adjacent pairs whose cosine similarity exceeds 0.7.
- 4) **Fully connected**: connect all distinct node pairs with cosine-similarity weights.

Figure 3 illustrates segment graphs. The recurrence matrices in Fig. 4 show the cosine similarities used to identify non-adjacent structural connections.

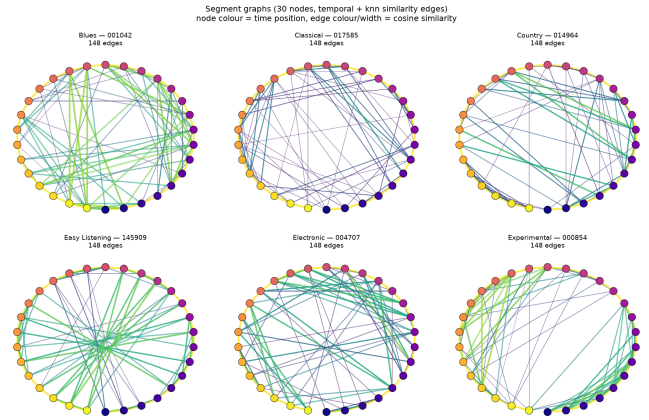


Fig. 3. Segment graphs across six genres, with 30 temporal nodes and temporal plus kNN acoustic-similarity edges.

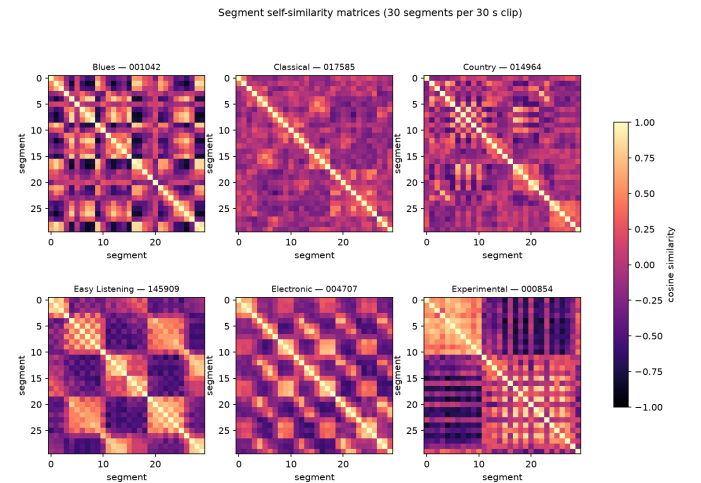


Fig. 4. Restored recurrence matrices with 30 segments per clip. Off-diagonal similarities identify candidate recurring structures used for similarity edges.

2) *Chord-transition graphs*: Chroma frames are compared with 24 normalized triad templates: 12 major templates with intervals $[0, 4, 7]$ and 12 minor templates with intervals $[0, 3, 7]$. Each frame receives its maximum-similarity chord assignment. The `graph_chord` variant is unsmoothed; `graph_chordsmooth` applies the reported 21-frame median filter (approximately 0.5 s). Smoothing is therefore an ablation, not common to both variants. The reported node-retention rule requires a chord to persist for at least three frames. Each retained chord node carries 37 features: a 24-dimensional one-hot identity, 12-dimensional mean chroma during activation, and one duration-share scalar. Directed edges connect consecutive distinct chords; transition counts are divided by the maximum observed transition count. These are relative weights, not row-normalized transition probabilities.

Figure 5 illustrates the resulting chord-transition graphs. Figure 6 summarizes segment-edge counts, distinct chord counts, and transition counts across 800 training tracks.

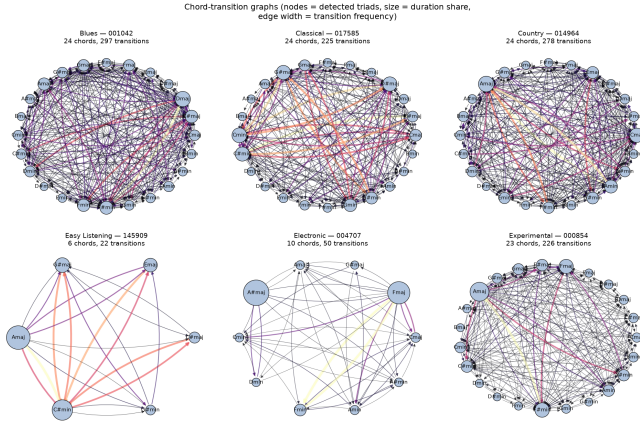


Fig. 5. Restored chord-transition graphs across six genres. Nodes are major/minor triads, node size represents duration share, and arrows represent observed transitions.

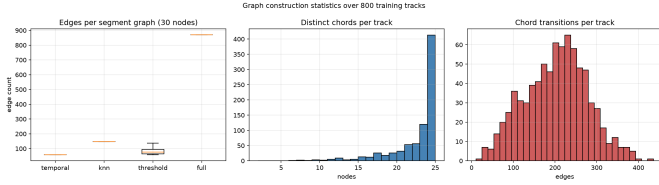


Fig. 6. Restored graph statistics over 800 training tracks: segment-edge counts by topology, distinct chord counts, and chord-transition counts.

B. GNN Models

The PyTorch Geometric implementation supports GAT [2], GraphSAGE [4], GCN, and GIN. Person 3’s report specifies *four* GAT attention heads, replacing the unsupported two-head description. Each convolution is followed by BatchNorm, ReLU, and dropout 0.30. Global mean pooling produces a 256-dimensional graph embedding; concatenated mean-plus-max pooling produces 512 dimensions. The selected model uses mean pooling. The stated readout implies 256-dimensional node representations immediately before pooling, but individual hidden-layer widths and per-head dimensions are not supplied and require saved configurations for verification.

The encoder-ablation GAT has 206,228 reported parameters; the combined final GAT has 208,020. These are distinct configurations, not interchangeable counts. The final model combines two layers, 60 nodes, 280-dimensional input features, and $k = 3$ kNN edges. The depth-sweep table does not identify its backbone, so its entries are labeled by layer count rather than “GAT layers.” Four heads are documented in the personal report, not independently checkpoint-verified.

C. Training and Controlled Evaluation

Pooled features or chord sequences are precomputed in 512-track chunks and cached on disk. Feature-scaling means and standard deviations are fitted *only on training data* and applied unchanged to validation and test graphs. PyTorch Geometric batches retain `track_id` attributes.

GNN TRAINING SETTINGS

Setting	Reported value or procedure
Objective	Binary cross-entropy with logits; 20 multi-label genres
Optimizer	AdamW
Learning rate / decay	0.001 / 0.0001
Schedule / duration	Cosine annealing; maximum 80 epochs
Gradient clipping	Norm 1.0
Early stopping	Validation AUC-PR; patience 15 epochs
Threshold tuning	Per-class validation search over 0.05–0.95; fixed for testing
Feature scaling	Fit on training features only
CNN replication	Same multi-label partitions, threshold procedure, and metrics

Per-class thresholds maximize class F1 on validation predictions, never test predictions. Their optimized class-wise F1 values are the components of Macro-F1. Grid spacing and training batch size are not specified in the contribution report. The controlled CNN replication imports Person 1’s exact four-layer architecture from `src/cnn_model.py`, using the same 128-bin mel inputs and evaluation protocol.

1) *Metric implementation:* Macro-F1 averages class F1 values equally; Micro-F1 pools label decisions. The per-genre plot explicitly reports average precision. However, the supplied PDFs do not establish whether aggregate “AUC-PR” uses average precision or integrated precision–recall area, or whether class averaging is macro, micro, or weighted. We retain the source label without asserting an implementation. The underlying `src/metrics.py` and evaluation outputs are required to resolve this. Person 3 reports a shared metric implementation for GNN and CNN; equivalence with BERT and fusion metrics is not established.

V. GNN EMPIRICAL ABLATIONS AND COMPARATIVE ANALYSIS

The contribution report describes 43 experimental runs across *five* axes: topology, node granularity, features, encoder, and depth. Table II restores all alternatives tabulated there, rather than enumerating all 43 configurations. Chord variants belong to topology, not a sixth axis. Scores are reported on the FMA top-20 multi-label test partition; run-level validation rankings, repetitions, and uncertainty estimates were not supplied.

A. Topology and Configuration Selection

Fully connected graphs, not kNN graphs, lead the isolated topology sweep on all three metrics: 0.3307/0.4391/0.3216 versus 0.3218/0.4235/0.3143. The final combined model nevertheless uses kNN. Sparse connectivity offers an efficiency trade-off: 148 rather than 870 edges in the 30-node comparison. This is a possible rationale, not a documented validation-based selection decision; neither that decision record nor a fully connected counterpart to the final 60-node model is supplied. kNN is therefore the *selected final topology*, not the topology-sweep winner.

Unsmoothed chord graphs outperform smoothed graphs (AUC-PR 0.2870 versus 0.2330), but both trail segment graphs. Smoothing may remove useful variation; the comparison does not establish a causal explanation.

TABLE II

REPORTED GNN ABLATION ALTERNATIVES ON FMA TOP-20 MULTI-LABEL GENRES. COMPARISONS ARE WITHIN EACH AXIS; THE COMBINED MODEL IS A SEPARATE RUN.

Axis	Variant	Configuration / size	Macro-F1	Micro-F1	AUC-PR
Topology	Fully connected	30 nodes; 870 edges	0.3307	0.4391	0.3216
	Threshold ($\tau = 0.7$)	30 nodes; 294 edges	0.3173	0.4298	0.3145
	Temporal + kNN ($k = 3$)	30 nodes; 148 edges	0.3218	0.4235	0.3143
	Temporal only	30 nodes; 58 edges	0.3107	0.4110	0.3131
	Chord, unsmoothed	24 nodes; 258 edges	0.2911	0.3970	0.2870
	Chord, smoothed	11 nodes; 45 edges	0.2500	0.3397	0.2330
Node granularity	60 nodes (~ 0.5 s)	298 average edges	0.3318	0.4660	0.3267
	30 nodes (~ 1.0 s)	148 average edges	0.3199	0.4361	0.3145
	15 nodes (~ 2.0 s)	73 average edges	0.3136	0.4294	0.3130
	6 nodes (~ 5.0 s)	28 average edges	0.3101	0.4031	0.3103
Node features	Mel + chroma, mean + std	280 dimensions	0.3390	0.4330	0.3416
	Mel mean	128 dimensions	0.3240	0.4088	0.3196
	Mel + chroma mean	140 dimensions	0.3194	0.4361	0.3140
	Chroma mean	12 dimensions	0.2707	0.3699	0.2594
Encoder	GAT	206,228 parameters	0.3211	0.4561	0.3236
	GraphSAGE	371,604 parameters	0.3204	0.4284	0.3133
	GCN	204,692 parameters	0.3182	0.4316	0.3111
	GIN	402,068 parameters	0.3017	0.3966	0.2954
Depth	2 layers	239,764 parameters	0.3283	0.4247	0.3305
	3 layers	371,604 parameters	0.3196	0.4502	0.3190
	4 layers	503,444 parameters	0.3194	0.4196	0.3164
Combined final	GAT; 60 nodes; kNN; 280-d; 2 layers	208,020 parameters	0.3544	0.4410	0.3673

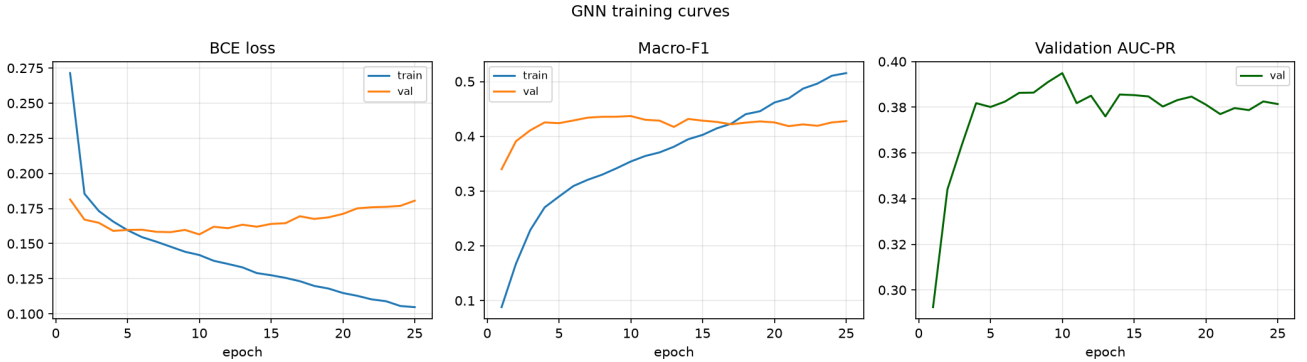


Fig. 7. Restored GNN learning curves: binary cross-entropy loss, Macro-F1, and validation AUC-PR. Validation loss eventually rises while training loss continues falling, indicating later overfitting rather than convergence “without overfitting.” Checkpoint selection and early stopping use validation AUC-PR.

B. Granularity, Features, Encoder, and Depth

Moving from 6 to 60 nodes raises Macro-F1 from 0.3101 to 0.3318 and AUC-PR from 0.3103 to 0.3267. Adding standard deviations to mel-plus-chroma means raises AUC-PR from 0.3140 to 0.3416. GAT leads the encoder comparison on AUC-PR (0.3236), with fewer parameters than GraphSAGE. Two layers yield the highest depth-sweep Macro-F1 and AUC-PR; three yield the highest Micro-F1. Weaker deeper-model AUC-PR is consistent with, but does not prove, over-smoothing.

C. Final Multi-Label and Single-Label Benchmarks

The selected final GAT achieves 0.3544 Macro-F1, 0.4410 Micro-F1, and 0.3673 AUC-PR, versus 0.4061, 0.5032, and 0.4260 for the controlled CNN. Reported final-model training time is approximately 35 seconds on an RTX 4090. Window

summary statistics may discard detail available to CNNs; the results do not establish GNN superiority. The final GNN’s AUC-PR is approximately $0.3673/0.0821 = 4.47$ times the label-prior baseline. This is *not* the effect of changing node granularity.

The following parity check is a different task: 16 mutually exclusive `genre_top` classes, trained with cross-entropy. Its scores must not be pooled with the top-20 multi-label benchmark.

SEPARATE 16-CLASS SINGLE-LABEL COMPARISON

Model	Macro-F1	Micro-F1	AUC-PR
Selected GAT	0.3341	0.6069	0.4022
CNN baseline	0.4209	0.6652	0.4814

Figure 7 restores the training curves with a corrected inter-

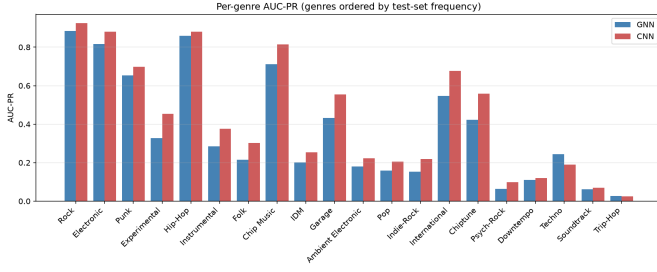


Fig. 8. Per-genre average precision comparing the selected GAT with the controlled CNN on the 20-genre task.

pretation. Figure 9 shows the standalone GNN embeddings alongside these results; it is separate from the fused projection and is qualitative, not a quantitative separability test.

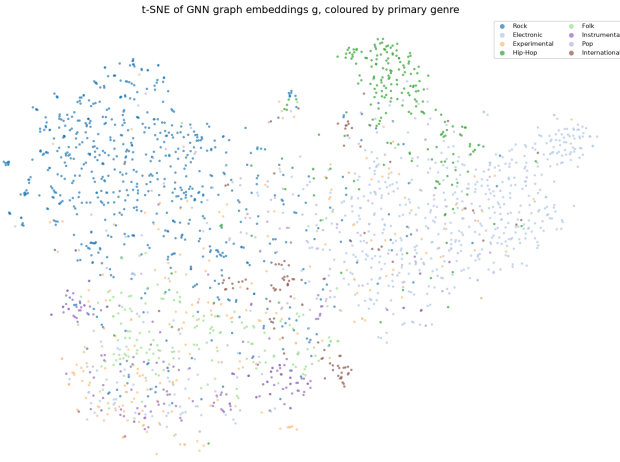


Fig. 9. Restored standalone GNN t-SNE of 256-dimensional embeddings across reported test tracks, colored by primary genre. This qualitative projection is distinct from the fused t-SNE in Fig. 12 and does not independently establish separability.

VI. TASK 3: GNN-BERT MULTIMODAL FUSION

Four reported strategies are `bert_only`, `gnn_only`, early concatenation $\mathbf{z} = [\mathbf{g}; \mathbf{t}]$, and multi-head cross-attention in which the BERT [CLS] query attends to GNN node embeddings. Table III and Fig. 10 retain the numerical comparison with the following qualifications.

A. Audio-Text Pairing and Evidence Status

Different encoder pretraining datasets do not themselves invalidate later fusion; the critical issue is the paired data used in Tasks 3 and 4. The supplied reports establish FMA graph construction and MusicCaps text training, but not the graph dataset used in those later experiments, each graph’s text source, or the matching procedure. Paired train/validation/test sizes, label vocabulary, and encoder transfer/freeze/retraining settings are also absent. No verified concrete audio-text pair can therefore be provided from these materials.

TABLE III
REPORTED FUSION COMPARISON (TASK 3). PAIRED DATASET, LABEL SPACE, AND CHECKPOINT PROVENANCE ARE UNVERIFIED.

Architecture	Params	Macro-F1	Micro-F1	AUC-PR
<code>bert_only</code>	109.6M	0.2656	0.2259	0.4226
<code>gnn_only</code>	70.9K	0.1048	0.1309	0.1199
<code>concat</code>	109.7M	0.1704	0.1589	0.3203
<code>Cross-attention</code>	110.2M	0.2168	0.2497	0.3181

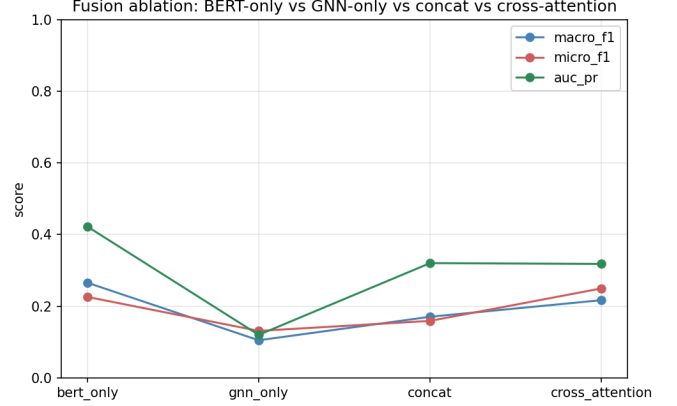


Fig. 10. Original reported Task 3 comparison; interpretation is conditional on verification of paired data and baseline configurations.

For example, FMA track 000181 occurs in the exported graph samples, but no corresponding caption or cross-dataset mapping is supplied. A `track_id` is a join key within an established dataset, not an automatic association between FMA recordings and MusicCaps videos. Metadata-derived text, if used, must be described and audited for target-label leakage. Neither metadata pairing nor expert-caption pairing is assumed here. Until pair manifests and evaluation records are supplied, Tasks 3 and 4 remain reported experiments with unverified provenance, not demonstrated semantic alignment.

B. Standalone Versus Fusion-Specific Baselines

The fusion `gnn_only` model has 70.9K parameters and Macro-F1 0.1048; it is not established as the standalone 208,020-parameter GAT with Macro-F1 0.3544. Its architecture and checkpoint are unspecified. Likewise, the fusion `bert_only` result is distinct from the 50-tag MusicCaps Task 1 result. Shared names or similar parameter counts do not establish identical data, labels, weights, or evaluation.

C. Qualified Comparison

Cross-attention improves Macro-F1 and Micro-F1 over concatenation by 0.0464 and 0.0908, but AUC-PR falls from 0.3203 to 0.3181. It also trails the fusion BERT-only model on Macro-F1 (0.2168 versus 0.2656) and AUC-PR (0.3181 versus 0.4226), while exceeding its Micro-F1. These results do not support uniform superiority or a causal explanation involving attention or gradient dilution.

TABLE IV
REPORTED CROSS-MODAL RETRIEVAL (TASK 4); MATCHING RULES AND
CANDIDATE-POOL SIZE ARE NOT SUPPLIED.

Direction	R@1	R@5	R@10	MRR
Text $\rightarrow$ audio graph	0.0160	0.0560	0.1280	0.0567
Audio graph $\rightarrow$ text	0.0200	0.0640	0.1400	0.0635

VII. TASK 4: CROSS-MODAL CONTRASTIVE ALIGNMENT (BONUS TASK)

The reported dual encoder projects graph and text representations into a normalized 128-dimensional space. For a batch of N putative matched pairs, let $\mathbf{g}_i, \mathbf{t}_i$ denote normalized projections and $\tau > 0$ the temperature. The symmetric InfoNCE-style objective [5] is split across lines to fit the two-column template:

$$\begin{aligned}\ell_i^{g \rightarrow t} &= -\log \frac{\exp(\mathbf{g}_i^\top \mathbf{t}_i / \tau)}{\sum_{j=1}^N \exp(\mathbf{g}_i^\top \mathbf{t}_j / \tau)}, \\ \ell_i^{t \rightarrow g} &= -\log \frac{\exp(\mathbf{t}_i^\top \mathbf{g}_i / \tau)}{\sum_{j=1}^N \exp(\mathbf{t}_i^\top \mathbf{g}_j / \tau)}, \\ \mathcal{L}_{\text{NCE}} &= \frac{1}{2N} \sum_{i=1}^N (\ell_i^{g \rightarrow t} + \ell_i^{t \rightarrow g}).\end{aligned}\quad (2)$$

A. Retrieval Protocol and Limitations

Table IV and Fig. 11 reproduce the reported retrieval values. The supplied reports do not define a correct match (exact recording, caption identity, or shared genre), candidate-pool size, query count, or treatment of multiple relevant items. They do not establish the held-out paired split or whether pools span the test set or individual batches. A verified retrieval or chance-level comparison is therefore unavailable. The original consolidated table’s unexplained random $R@5 = 0.0200$ is not assigned to the unrelated genre-prior baseline.

The original report also gives prompt-based classification scores using “A {genre} music track.”: 0.1747 Macro-F1, 0.1402 Micro-F1, and 0.2136 AUC-PR. These are retained as *reported prompt-based scores*; missing vocabulary, training-exposure, pairing, and threshold details preclude a verified zero-shot claim.

VIII. MASTER EXPERIMENTAL RESULTS

Table V separates datasets and label spaces rather than implying one ranking. Only the FMA top-20 random, CNN, and standalone GAT rows form the documented same-task comparison. Task 1 addresses MusicCaps caption tagging; Tasks 3 and 4 lack paired-data records establishing comparability. The 16-class task remains separate in Section V.

IX. QUALITATIVE CASE STUDIES AND CROSS-MODAL ALIGNMENT

All three examples retained in Table VI are labeled Rock, and two share the artist Ariel Pink. They are Rock-focused illustrations, not “three diverse test tracks.” Test-split membership is not verified. Confidence values are retained as

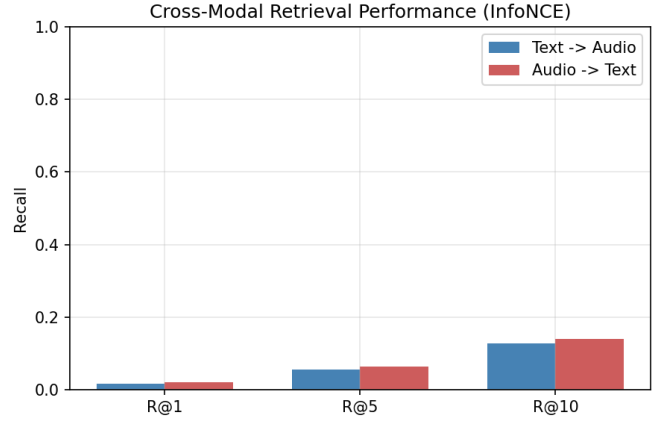


Fig. 11. Original reported retrieval Recall@K under symmetric InfoNCE. The evaluation protocol remains unverified.

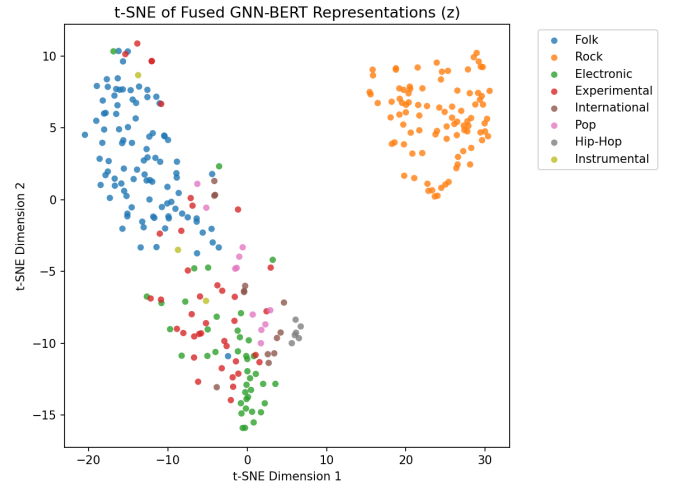


Fig. 12. Original t-SNE of reported fused representations, colored by top genre. This is not the standalone GNN projection; paired-data provenance remains unverified.

reported predictions, while detailed attention-to-chorus and lyrical-alignment claims are not treated as established without corresponding captions, attention outputs, and audio annotations. The original fused t-SNE (Fig. 12) alone does not demonstrate cross-modal alignment.

X. TEAM WORK BREAKDOWN AND RESPONSIBILITIES

The reported division of work is as follows:

- **Person 1:** Original audio preprocessing design, 22.05 kHz resampling and 128-mel/12-chroma specifications, FMA partitioning, and CNN baseline architecture.
- **Person 2:** MusicCaps preparation, aspect-tag vocabulary, BERT tokenization and fine-tuning, multi-label classification, evaluation, and training/validation visualizations.
- **Person 3 (Azra Humayra Alam Prova):** GPU feature extraction and CPU/GPU validation; float16 memory-mapped storage redesign; corruption handling; multi-label/single-label target preparation; artist-leakage au-

TABLE V
CONSOLIDATED REPORTED RESULTS BY DATASET AND LABEL SPACE. SCORES ACROSS DIFFERENT TASKS ARE NOT DIRECTLY COMPARABLE.

Model / task	Dataset	Label space	Macro-F1	Micro-F1	AUC-PR	R@5
Random tags (B1)	FMA-medium	Top-20 multi-label genres	0.0977	0.2599	0.0821	—
CNN mel-spec (B2)	FMA-medium	Top-20 multi-label genres	0.4061	0.5032	0.4260	—
Task 2: GAT	FMA-medium	Top-20 multi-label genres	0.3544	0.4410	0.3673	—
Task 1: BERT (B3) ^a	MusicCaps	50 multi-label aspect tags	0.6126	0.7232	0.7130	—
Task 3: cross-attention ^b	Paired data unspecified	Not reported	0.2168	0.2497	0.3181	—
Task 4: dual encoder ^{b,c}	Paired data unspecified	Prompt genres unspecified	0.1747	0.1402	0.2136	0.0560

^aHeld-out MusicCaps test scores (829 clips); evaluation and validation-based checkpoint selection are described in Section III-C.

^bRetained as reported, not as verified aligned-data benchmarks.

^cClassification columns concern prompt-based classification; R@5 concerns text-to-audio retrieval, a separate evaluation.

TABLE VI
REPORTED ROCK-FOCUSED EXAMPLES; SPLIT MEMBERSHIP, PAIRING, AND ATTENTION INTERPRETATIONS ARE UNVERIFIED.

Track / artist	Reported graph	Reported true / predicted
000181, "Gopacapulco," Ariel Pink	60 nodes, 298 edges	Rock / Rock (92.4%)
000182, "Jules Lost His Jewels," Ariel Pink	60 nodes, 298 edges	Rock / Rock (88.1%)
000564, "The Sugar Society," Cinwaves	60 nodes, 298 edges	Rock / Rock (86.7%)

ditioning and compatible split enrichment; segment/chord graph construction; GAT, GraphSAGE, GCN, and GIN encoders; caching, training-only scaling, training, early stopping, and validation-only threshold tuning; 43 reported runs across five ablation axes; controlled CNN replication and both task comparisons; graph visualizations, 24 sample exports, checkpoints, encoding interface, and integration handover.

- **Person 4:** Reported fusion architectures, paired-data integration work, fusion ablations, fused t-SNE, and contrastive dual-encoder retrieval. Pairing and evaluation provenance still require documentation as discussed in Sections VI–VII.
- **Joint contributions:** Experimental design, demo notebook preparation, and final academic paper compilation.

A. Implementation Deliverables and Handover

Person 3 reports 24 sample graphs across 12 genres, two per genre: Rock, International, Soul-RnB, Experimental, Folk, Blues, Electronic, Old-Time / Historic, Instrumental, Pop, Hip-Hop, and Jazz. Exports are described as PyTorch Geometric .pt and JSON files in `data/processed/graph_samples/`, indexed by `index.json`. JSON records contain `track_id`, labels, node/edge counts, feature dimensionality, `edge_index`, and edge weights.

Checkpoints and final results are reported under `results/final/`, with `results/RESULTS.md` and `person_3_handover.md` documenting results and integration. The `model.encode(batch)` interface returns

mean-pooled embeddings in $\mathbb{R}^{B \times 256}$. Attached track IDs support deterministic joins *when actual corresponding text exists*; this contract does not prove that FMA and MusicCaps were paired. These implementation artifacts are described in the contribution report but were not supplied with the PDFs.

XI. DISCUSSION, LIMITATIONS, AND CONCLUSION

A. Key Insights

- 1) **Complementary representations:** The controlled CNN outperforms the GNN on FMA top-20 genres (0.4061 versus 0.3544 Macro-F1). Graphs explicitly encode temporal recurrence, but adding text is not established to close this performance gap.
- 2) **Granularity and topology:** Moving from 6 to 60 nodes improves Macro-F1 by 0.0217 and AUC-PR by 0.0164. The separate $4.47\times$ AUC-PR ratio compares the final GNN with the label prior. Fully connected graphs win the topology sweep; kNN is retained in the combined model without proof of global optimality.
- 3) **Qualified fusion evidence:** Cross-attention improves both F1 measures over concatenation, but not AUC-PR, and does not uniformly outperform its BERT-only baseline. Missing pairing records preclude verified alignment and zero-shot conclusions.

B. Limitations and Future Directions

The central multimodal limitation is not simply different encoder datasets, but undocumented corresponding audio–text examples for later experiments. A reproducible continuation requires pair manifests, one inspectable matched example, paired split sizes, vocabulary, encoder settings, and retrieval relevance and candidate-pool definitions. Constructing MusicCaps audio graphs with their corresponding captions offers one route to an explicitly paired corpus; any metadata-derived alternative must be described and audited for label leakage.

Other gaps include filtered FMA validation/test counts, saved GAT settings, depth-sweep backbone, aggregate AUC-PR implementation, and final-configuration selection records. The reported 43 runs lack complete configurations and repeated-seed uncertainty estimates. Test-score ablation tables without a final selection trail do not prove validation-only

architecture selection. Rising later GNN validation loss also warrants caution about overfitting.

Full similarity computation scales quadratically with node count. Hierarchical graphs and beat-synchronous representations are future directions, alongside artist-disjoint evaluation, training-only scaling, and validation-only threshold tuning.

C. Conclusion

This report separates documented standalone results from multimodal claims awaiting verification. Person 3’s GPU extraction, storage redesign, corruption handling, leakage audit, graph encoders, controlled training, and five-axis ablations extend the group pipeline. The selected GAT achieves 0.3544 Macro-F1 and 0.3673 AUC-PR on FMA top-20 multi-label genres, below the CNN’s 0.4061 and 0.4260. The separate MusicCaps BERT baseline achieves held-out test scores of 0.6126 Macro-F1, 0.7232 Micro-F1, and 0.7130 AUC-PR on

829 clips. Fusion and contrastive architectures support further study, but current documentation does not establish verified paired-data retrieval or cross-modal alignment.

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